

Gluconeogenesis

Definition | Location | Pathway | Precursors | Energetics | Regulation | Clinical importance

Exam orientation

Write gluconeogenesis as the synthesis of glucose from non-carbohydrate precursors. Focus on the three bypass reactions, energy cost, Cori cycle, glucose-alanine cycle, regulation by glucagon/insulin, and alcohol-induced hypoglycemia.

1. Definition and basic idea

Definition: Gluconeogenesis is the formation of glucose from non-carbohydrate precursors such as lactate, pyruvate, glucogenic amino acids, glycerol and propionate.

Main purpose: It maintains blood glucose during fasting, starvation, prolonged exercise and low-carbohydrate intake, especially for glucose-dependent tissues such as brain, erythrocytes, renal medulla and testes.

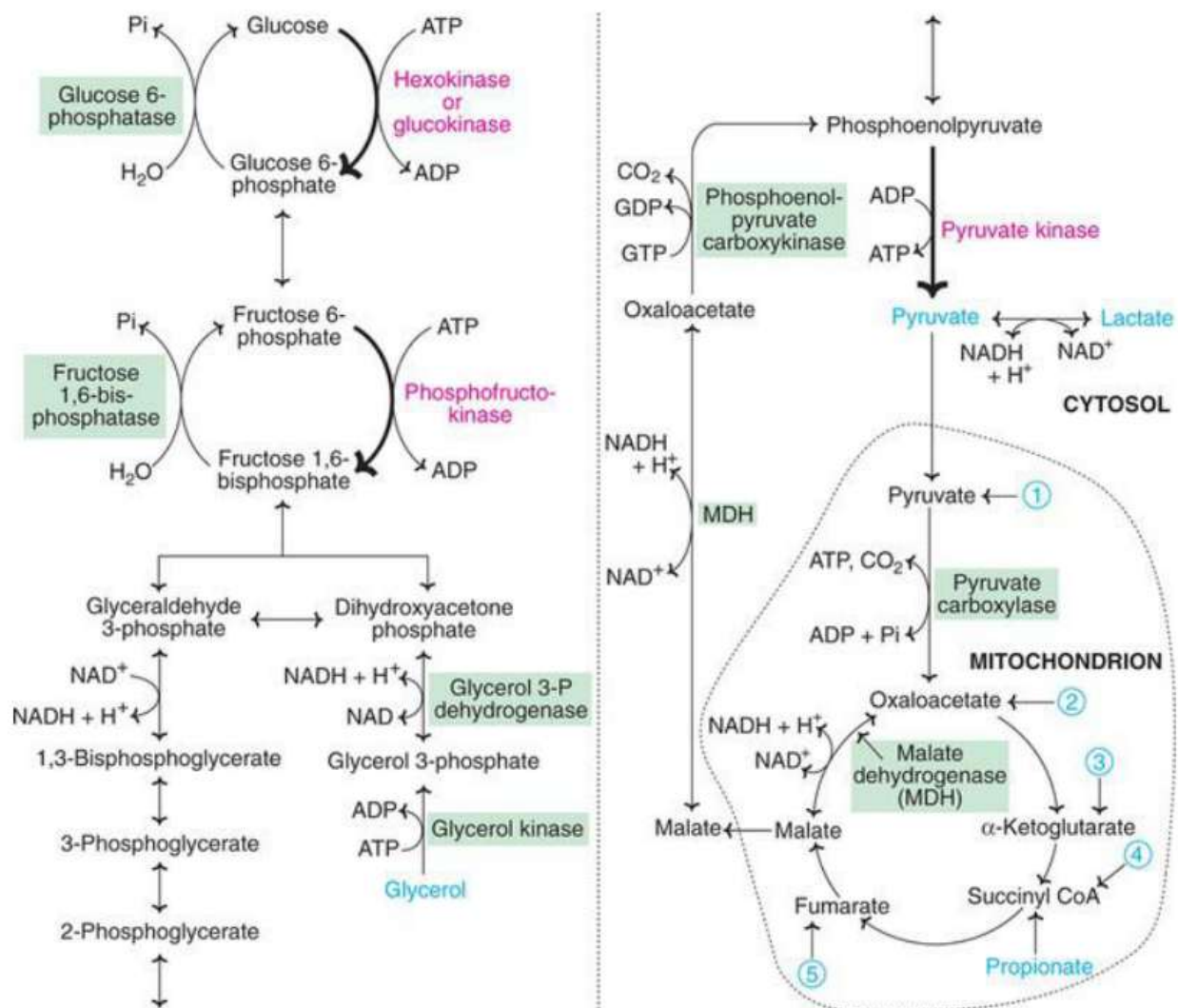
Not simply reverse glycolysis: Seven glycolytic reactions are reversible and can run backward. Three glycolytic reactions are irreversible, so gluconeogenesis uses special bypass enzymes.

Feature	High-yield point
Major organs	Liver is the main organ; kidney contributes especially in prolonged fasting.
Cellular location	Mostly cytosol; first bypass begins in mitochondria. Glucose-6-phosphatase is associated with endoplasmic reticulum.
Major precursors	Lactate, alanine and other glucogenic amino acids, glycerol, propionate, pyruvate.
Key bypass enzymes	Pyruvate carboxylase, PEP carboxykinase, fructose-1,6-bisphosphatase, glucose-6-phosphatase.
Energy requirement	For 1 glucose from 2 pyruvate: 4 ATP + 2 GTP + 2 NADH are used.
Hormonal control	Glucagon and cortisol favour it; insulin inhibits it.

2. Importance of gluconeogenesis

- Maintains blood glucose between meals and during fasting when dietary glucose is unavailable.
- Supplies glucose to brain and central nervous system; brain needs a continuous glucose supply in early fasting.
- Supplies glucose to erythrocytes, which lack mitochondria and depend completely on glycolysis.
- Clears lactate produced by active skeletal muscle and RBCs, preventing excessive lactate accumulation.
- Converts carbon skeletons of glucogenic amino acids into glucose during starvation or uncontrolled diabetes.
- Replenishes glucose for muscle through the Cori cycle and glucose-alanine cycle.

3. Overall pathway - source diagram



Source diagram: Pathway of gluconeogenesis showing bypass enzymes, precursor entry points and mitochondrial/cytosolic phases.

How to read the diagram

Start from pyruvate or lactate at the lower/right side. Pyruvate first enters mitochondria and becomes oxaloacetate. Oxaloacetate is converted to phosphoenolpyruvate (PEP), then the pathway proceeds upward through reversible glycolytic reactions until fructose-1,6-bisphosphate. The final two bypasses produce fructose-6-phosphate, glucose-6-phosphate and finally free glucose.

4. The three irreversible glycolytic steps and their gluconeogenic bypasses

Irreversible glycolysis step	Why it needs a bypass	Gluconeogenic bypass enzyme(s)	Main product
Glucose → Glucose-6-phosphate Hexokinase / glucokinase	Consumes ATP and is strongly irreversible.	Glucose-6-phosphatase	Free glucose
Fructose-6-phosphate → Fructose-1,6-bisphosphate Phosphofructokinase-1	Committed, rate-limiting glycolytic step.	Fructose-1,6-bisphosphatase	Fructose-6-phosphate
PEP → Pyruvate Pyruvate kinase	Produces ATP and is irreversible.	Pyruvate carboxylase + PEP carboxykinase	Phosphoenolpyruvate

5. Step-by-step pathway explained

Step 1: Pyruvate → Oxaloacetate → Phosphoenolpyruvate

- Pyruvate enters mitochondria and is carboxylated to oxaloacetate by pyruvate carboxylase.
- Pyruvate carboxylase requires biotin, ATP and CO₂, and is activated by acetyl-CoA.
- Oxaloacetate cannot freely cross the inner mitochondrial membrane, so it is commonly reduced to malate, transported to cytosol, and reoxidized to oxaloacetate.
- Cytosolic oxaloacetate is converted to phosphoenolpyruvate by phosphoenolpyruvate carboxykinase (PEPCK), using GTP and releasing CO₂.

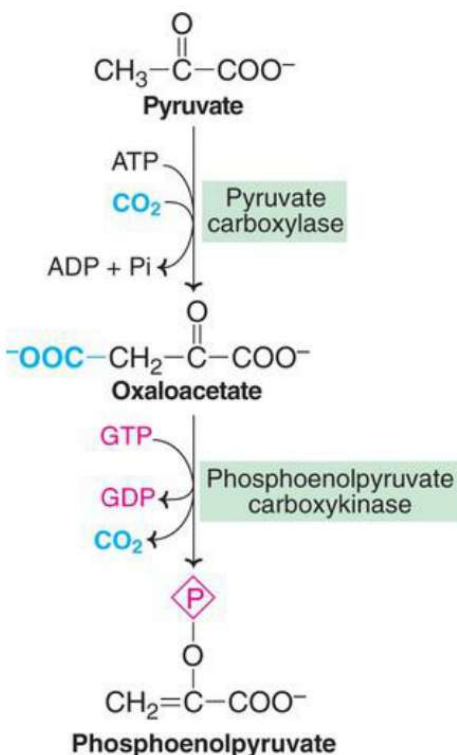


FIG. 13.12 Conversion of pyruvate to phosphoenolpyruvate.

Source diagram: Conversion of pyruvate to phosphoenolpyruvate through oxaloacetate.

Step 2: PEP → Fructose-1,6-bisphosphate

After PEP formation, the pathway mostly follows the reverse of glycolysis through a series of reversible reactions:

- PEP → 2-phosphoglycerate → 3-phosphoglycerate → 1,3-bisphosphoglycerate → glyceraldehyde-3-phosphate.
- Glyceraldehyde-3-phosphate and dihydroxyacetone phosphate combine to form fructose-1,6-bisphosphate.
- NADH is required at the glyceraldehyde-3-phosphate dehydrogenase step.

Step 3: Fructose-1,6-bisphosphate → Fructose-6-phosphate

Fructose-1,6-bisphosphatase hydrolyses fructose-1,6-bisphosphate to fructose-6-phosphate. This is a key regulatory step of gluconeogenesis and requires Mg²⁺.

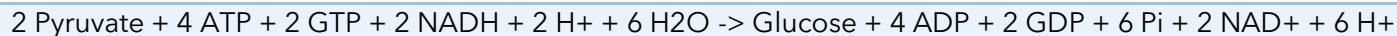
- Activated by: citrate and low-energy/fasting conditions.
- Inhibited by: AMP and fructose-2,6-bisphosphate.

Step 4: Fructose-6-phosphate → Glucose-6-phosphate → Glucose

- Fructose-6-phosphate is isomerized to glucose-6-phosphate by phosphoglucose isomerase.
- Glucose-6-phosphatase hydrolyses glucose-6-phosphate to free glucose and inorganic phosphate.
- Glucose-6-phosphatase is present mainly in liver and kidney, but absent in muscle, brain and adipose tissue. Therefore, muscle glycogen cannot directly maintain blood glucose.

6. Energetics of gluconeogenesis

Overall reaction from pyruvate



Reaction/phase	Energy used	Reason
2 Pyruvate $\rightarrow$ 2 Oxaloacetate	2 ATP	Pyruvate carboxylase reaction.
2 Oxaloacetate $\rightarrow$ 2 PEP	2 GTP	PEP carboxykinase reaction.
2 3-phosphoglycerate $\rightarrow$ 2 1,3-bisphosphoglycerate	2 ATP	Reverse of phosphoglycerate kinase step.
2 1,3-bisphosphoglycerate $\rightarrow$ 2 Glyceraldehyde-3-phosphate	2 NADH	Reverse of GAP dehydrogenase step.
Total cost per glucose from 2 pyruvate	4 ATP + 2 GTP + 2 NADH	A costly pathway, used only when glucose production is essential.

Important comparison: glycolysis of glucose to 2 pyruvate gives a net gain of 2 ATP and 2 NADH, but gluconeogenesis from 2 pyruvate consumes 4 ATP, 2 GTP and 2 NADH. This prevents a futile cycle and makes regulation essential.

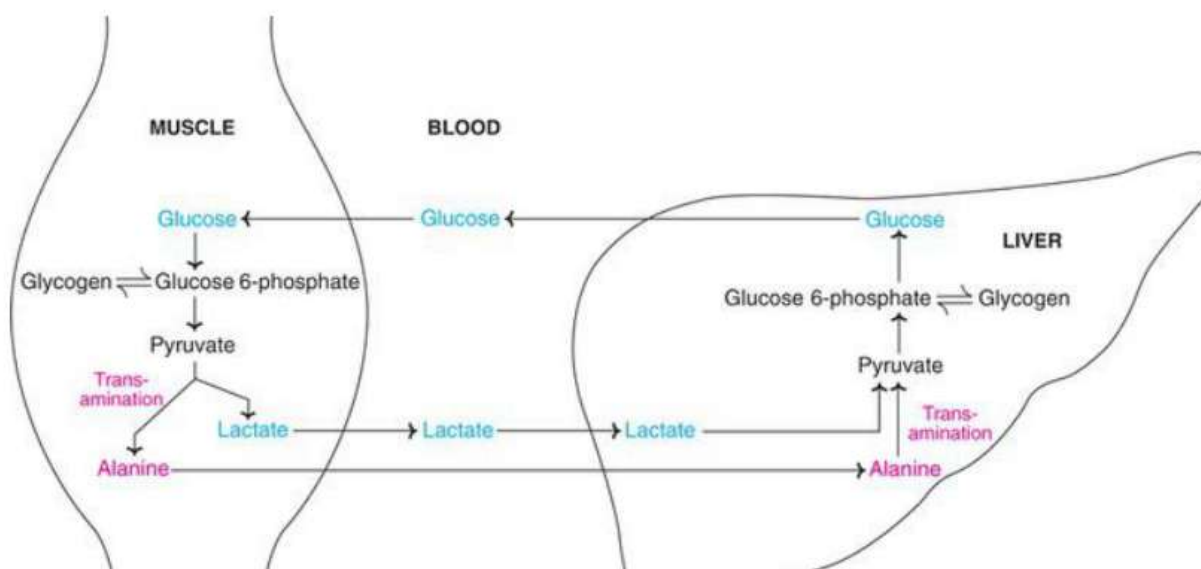
7. Precursors and entry points

Precursor	Source	Entry into gluconeogenesis	Key point
Lactate	Active skeletal muscle and RBCs	Lactate $\rightarrow$ Pyruvate by lactate dehydrogenase	Forms the Cori cycle.
Alanine	Skeletal muscle protein metabolism	Alanine $\rightarrow$ Pyruvate by transamination	Carries amino nitrogen to liver.
Glucogenic amino acids	Muscle protein during fasting/starvation	Enter as pyruvate or TCA intermediates	All except leucine and lysine are at least partly glucogenic.
Glycerol	Breakdown of triacylglycerol in adipose tissue	Glycerol $\rightarrow$ Glycerol-3-phosphate $\rightarrow$ DHAP	Uses glycerol kinase in liver and kidney.
Propionate	Odd-chain fatty acids and some amino acids	Propionyl-CoA $\rightarrow$ Succinyl-CoA $\rightarrow$ TCA intermediates	Requires biotin and vitamin B12-dependent steps.

Detailed precursor notes

- Lactate is the most important precursor during intense exercise and in red blood cells; it is reconverted to pyruvate in the liver.
- Alanine is important during fasting because it transports both carbon skeleton and amino nitrogen from muscle to liver.
- Glycerol comes from adipose tissue lipolysis. It enters at DHAP after phosphorylation and oxidation.
- Propionate gives net glucose because it enters through succinyl-CoA, unlike even-chain fatty acids that form only acetyl-CoA.

Cori cycle and glucose-alanine cycle



Source diagram: Cori cycle and glucose-alanine cycle between muscle, blood and liver.

Cori cycle: In exercising muscle, pyruvate is reduced to lactate to regenerate NAD⁺. Lactate travels to liver, becomes pyruvate again, and is converted to glucose by gluconeogenesis. Glucose returns to muscle for energy.

Glucose-alanine cycle: In muscle, pyruvate accepts an amino group to become alanine. Alanine travels to liver, is converted back to pyruvate, and the carbon skeleton is used for gluconeogenesis while nitrogen is disposed through urea formation.

8. Regulation of gluconeogenesis

Regulator	Effect on gluconeogenesis	Mechanism / meaning
Glucagon	Increases	Activates cAMP pathway; inhibits pyruvate kinase; lowers fructose-2,6-bisphosphate, which favours fructose-1,6-bisphosphatase.
Insulin	Decreases	Promotes glycolysis and glycogenesis; suppresses gluconeogenic enzyme expression.
Acetyl-CoA	Increases	Allosterically activates pyruvate carboxylase during fasting/fatty acid oxidation.
AMP	Decreases	Signals low energy; inhibits fructose-1,6-bisphosphatase.
Fructose-2,6-bisphosphate	Decreases	Inhibits fructose-1,6-bisphosphatase and activates PFK-1, promoting glycolysis.
Substrate availability	Increases when high	More lactate, alanine, glycerol and glucogenic amino acids increase glucose production.
Cortisol	Increases	Mobilizes amino acids and supports gluconeogenic enzyme synthesis during stress/starvation.

Fed state vs fasting state

State	Hormonal pattern	Dominant liver metabolism	Effect on blood glucose
Fed state	High insulin, low glucagon	Glycolysis and glycogenesis increase; gluconeogenesis decreases.	Excess glucose is stored or used.
Early fasting	Glucagon rises	Glycogenolysis first, then gluconeogenesis increases.	Blood glucose is maintained.
Prolonged fasting	Low insulin, high glucagon/cortisol	Gluconeogenesis becomes the major glucose source; kidney contribution rises.	Glucose is spared for dependent tissues.

9. Clinical and applied importance

Alcohol inhibits gluconeogenesis

Ethanol metabolism produces excess NADH in liver. High NADH shifts pyruvate toward lactate and oxaloacetate toward malate, so the main substrates for gluconeogenesis become unavailable. This can cause hypoglycemia, especially in fasting, malnourished or insulin-treated diabetic patients.

Reaction logic in alcohol intake

Ethanol → Acetaldehyde produces NADH. Excess NADH drives: Pyruvate → Lactate and Oxaloacetate → Malate. Therefore pyruvate and oxaloacetate are pulled away from glucose synthesis.

Why most fat cannot become glucose

Even-chain fatty acids are degraded to acetyl-CoA. Acetyl-CoA cannot be converted back to pyruvate in humans, and its two carbons are lost as CO₂ in the citric acid cycle. Therefore, most fatty acids cannot provide net glucose. Exceptions related to fat metabolism include glycerol from triacylglycerol and propionate from odd-chain fatty acids.

Diabetes and starvation

- In uncontrolled diabetes mellitus, low effective insulin and high glucagon increase gluconeogenesis, contributing to hyperglycemia.
- During starvation, amino acids from muscle and glycerol from adipose tissue become important glucose precursors.

- In prolonged fasting, the brain adapts partly to ketone bodies, reducing but not eliminating the need for glucose.

10. Exam-ready summary

Question cue	Answer core
Define gluconeogenesis	Synthesis of glucose from non-carbohydrate precursors.
Main site	Liver mainly; kidney also, especially during prolonged fasting.
Major precursors	Lactate, alanine/glucogenic amino acids, glycerol, propionate, pyruvate.
Bypass enzymes	Pyruvate carboxylase, PEPCK, fructose-1,6-bisphosphatase, glucose-6-phosphatase.
Energy cost	4 ATP + 2 GTP + 2 NADH per glucose from 2 pyruvate.
Cori cycle	Muscle/RBC lactate → liver glucose → back to muscle/blood.
Regulation	Glucagon, acetyl-CoA and substrate availability stimulate; insulin, AMP and F-2,6-BP inhibit.
Alcohol effect	Inhibits gluconeogenesis by increasing hepatic NADH and causes hypoglycemia risk.

Suggested 10-mark answer structure

- Definition and purpose: glucose production from non-carbohydrate precursors to maintain blood glucose.
- Site: mainly liver, partly kidney; mitochondrial, cytosolic and ER-associated steps.
- Precursors: lactate, alanine/glucogenic amino acids, glycerol, propionate and pyruvate.
- Pathway: explain three bypasses of irreversible glycolytic steps.
- Energetics: mention use of 4 ATP, 2 GTP and 2 NADH per glucose.
- Regulation: glucagon stimulates; insulin inhibits; acetyl-CoA activates pyruvate carboxylase.
- Clinical importance: fasting, diabetes, alcohol-induced hypoglycemia, Cori cycle.

One-line memory aid

Remember

Gluconeogenesis is "glucose rescue" during fasting: Liver uses Lactate, Alanine, Glycerol and other glucogenic carbons, spends 6 high-energy phosphate bonds, and is switched ON by glucagon.

Glycolysis vs gluconeogenesis - quick comparison

Point	Glycolysis	Gluconeogenesis
Direction	Glucose → Pyruvate/Lactate	Pyruvate/Lactate → Glucose
Function	Energy production	Blood glucose maintenance
Energy	Produces ATP and NADH	Consumes ATP, GTP and NADH
Major site	Most cells; high in muscle/RBCs	Mainly liver and kidney
Hormonal state	Fed state; insulin favours	Fasting state; glucagon favours

Common mistakes to avoid

- Do not write gluconeogenesis as the exact reverse of glycolysis; remember the three bypasses.
- Do not write muscle as a major organ for releasing glucose into blood; muscle lacks glucose-6-phosphatase.
- Do not say acetyl-CoA gives net glucose in humans; even-chain fatty acids cannot be converted into net glucose.
- Do not forget energy cost: 4 ATP + 2 GTP + 2 NADH per glucose from 2 pyruvate.

Quick viva / short-answer practice

Question	Answer
Name the rate-limiting enzyme.	Fructose-1,6-bisphosphatase is the key regulatory enzyme.
Which enzyme needs biotin?	Pyruvate carboxylase.
Which step uses GTP?	Oxaloacetate → phosphoenolpyruvate by PEPCK.
Why cannot muscle release free glucose?	Muscle lacks glucose-6-phosphatase.
Which hormone switches ON gluconeogenesis?	Glucagon; cortisol also supports it during stress/starvation.
Which metabolite activates pyruvate carboxylase?	Acetyl-CoA.
Why does alcohol cause hypoglycemia?	It raises hepatic NADH, diverting pyruvate to lactate and oxaloacetate to malate.

*Source used: Uploaded PDF - Biochemistry, U. Satyanarayana & U. Chakrapani, 4th Edition, Chapter 13: Metabolism of Carbohydrates.
Diagrams were taken from the provided source and resized for readability.*